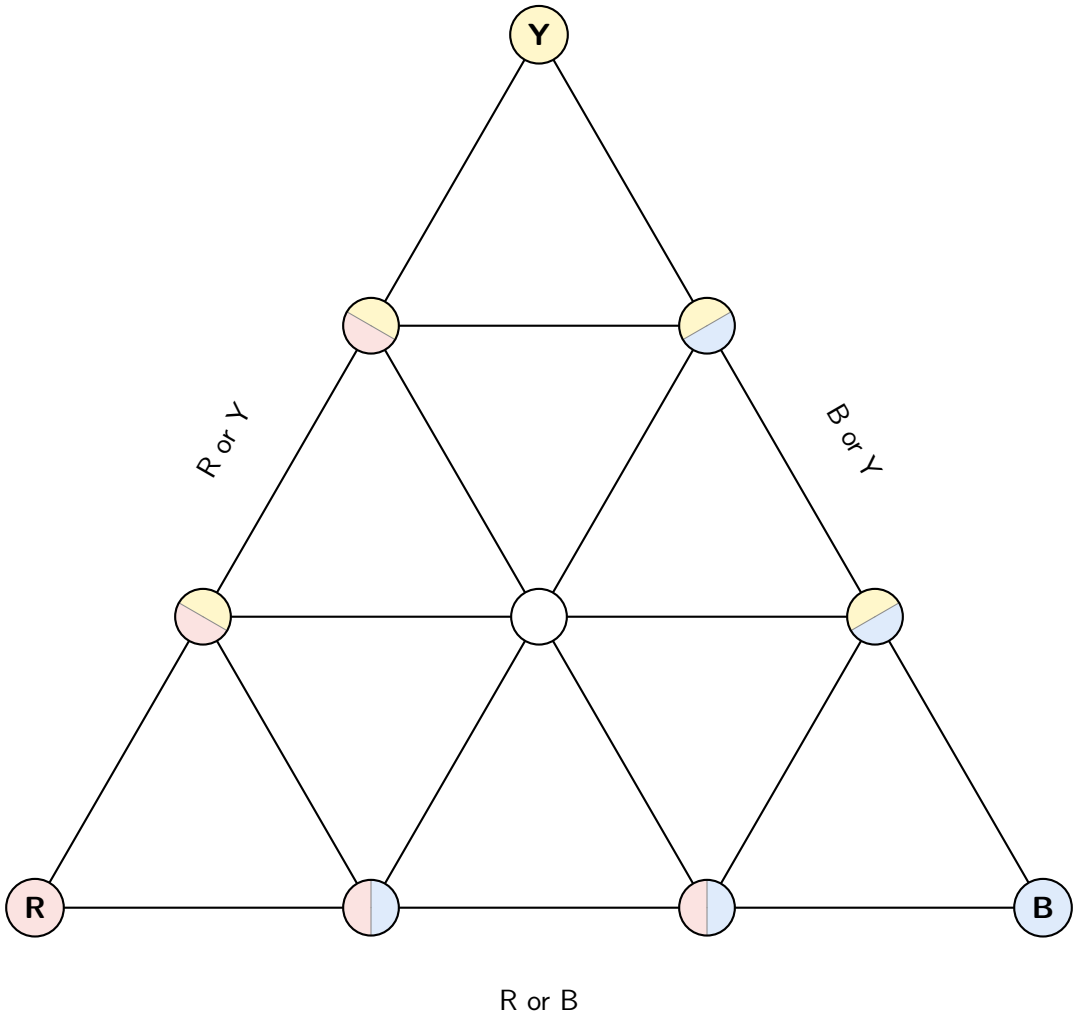
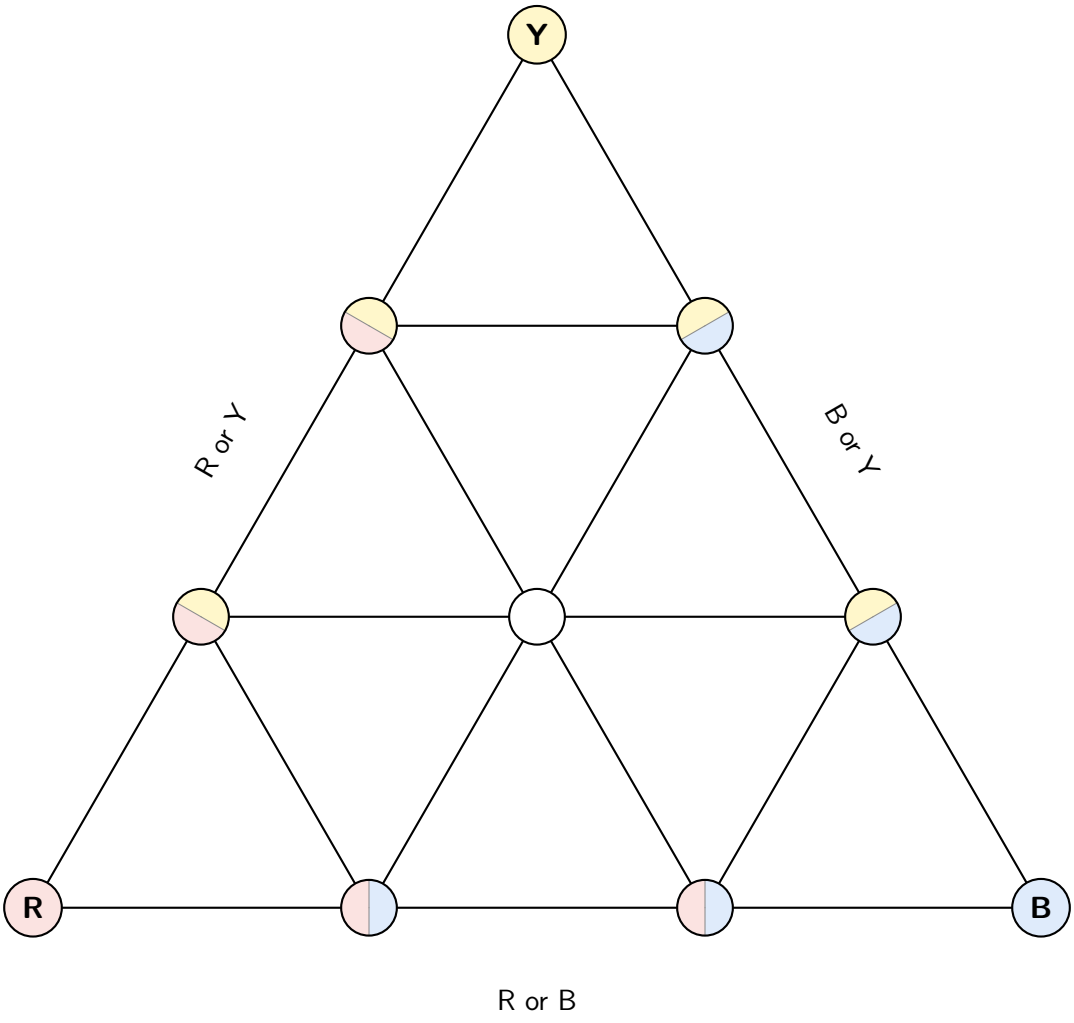
Put R, B, or Y on every empty dot. Keep the printed letters. Each outside side allows only the letters beside it. A little triangle has no lines inside.

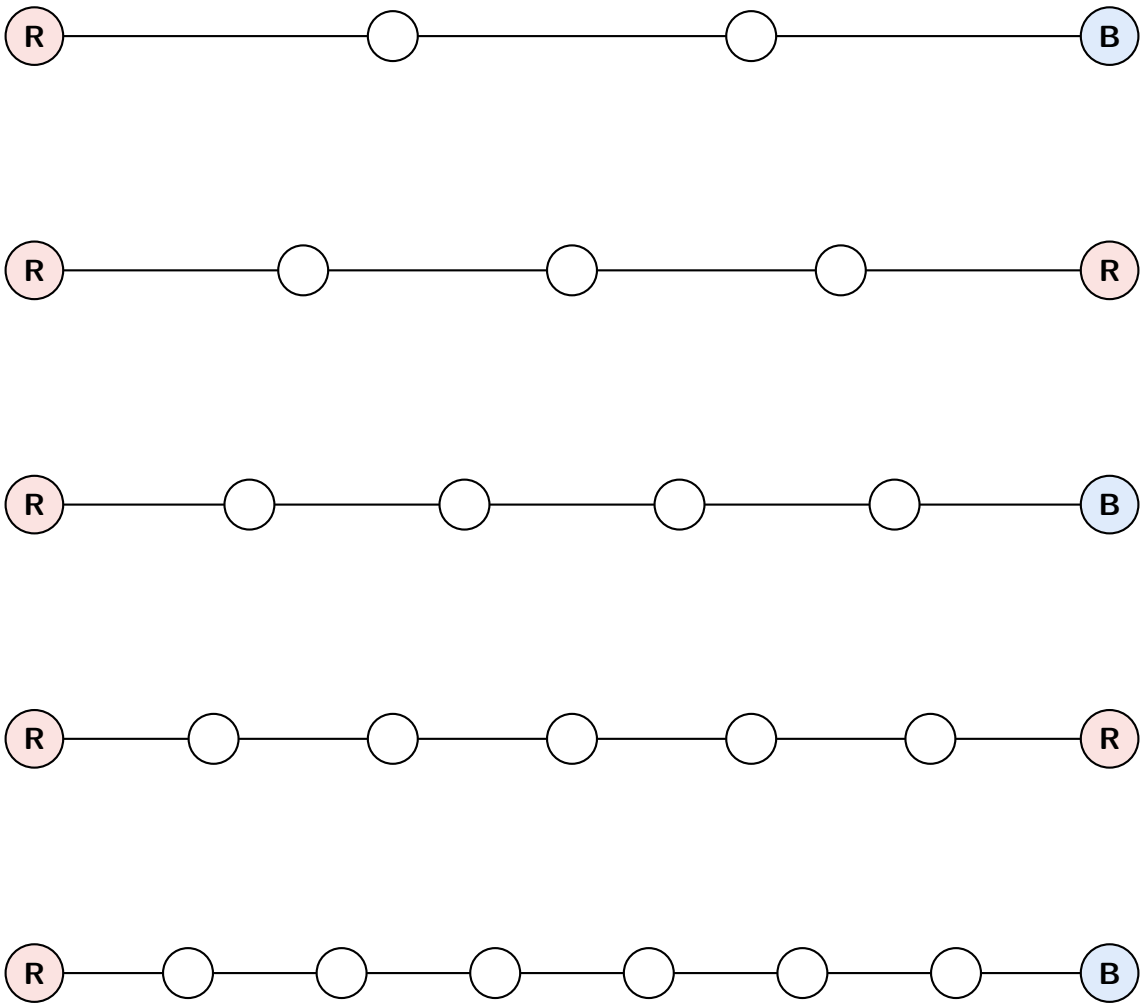
Problem 1: Try fillings with R, B, and Y counters. Find a filling with as few little triangles containing R, B, and Y as you can, and another with as many as you can. Write each record on its own board. Which counts between your two records can you make?



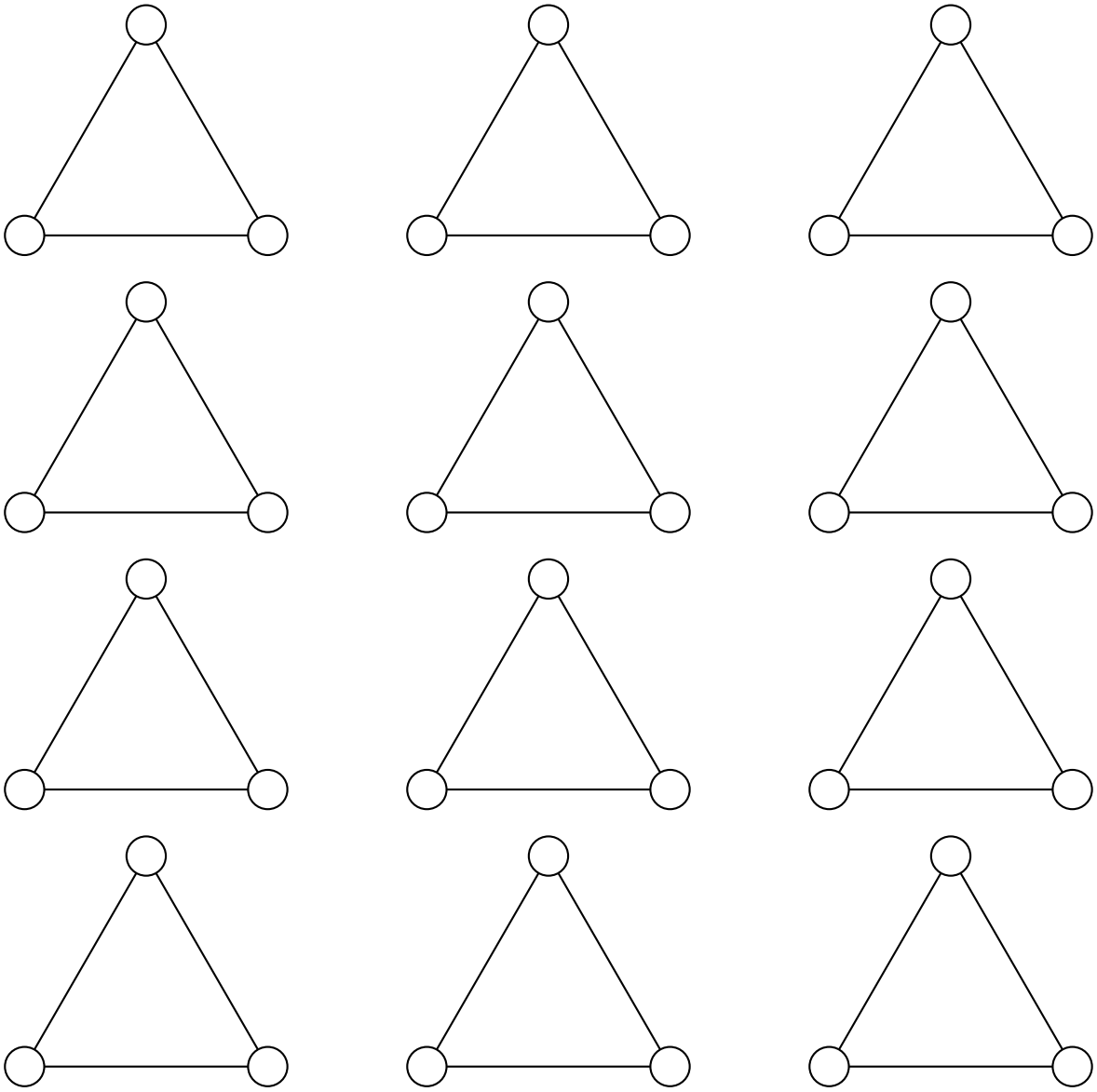
Another board for Problem 1.



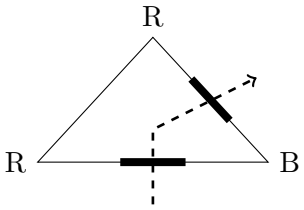
Problem 2: An edge with R at one end and B at the other is a door. Write R or B on every empty dot and mark the doors. Which numbers of doors can each row have? Explain your answer for rows of any length.



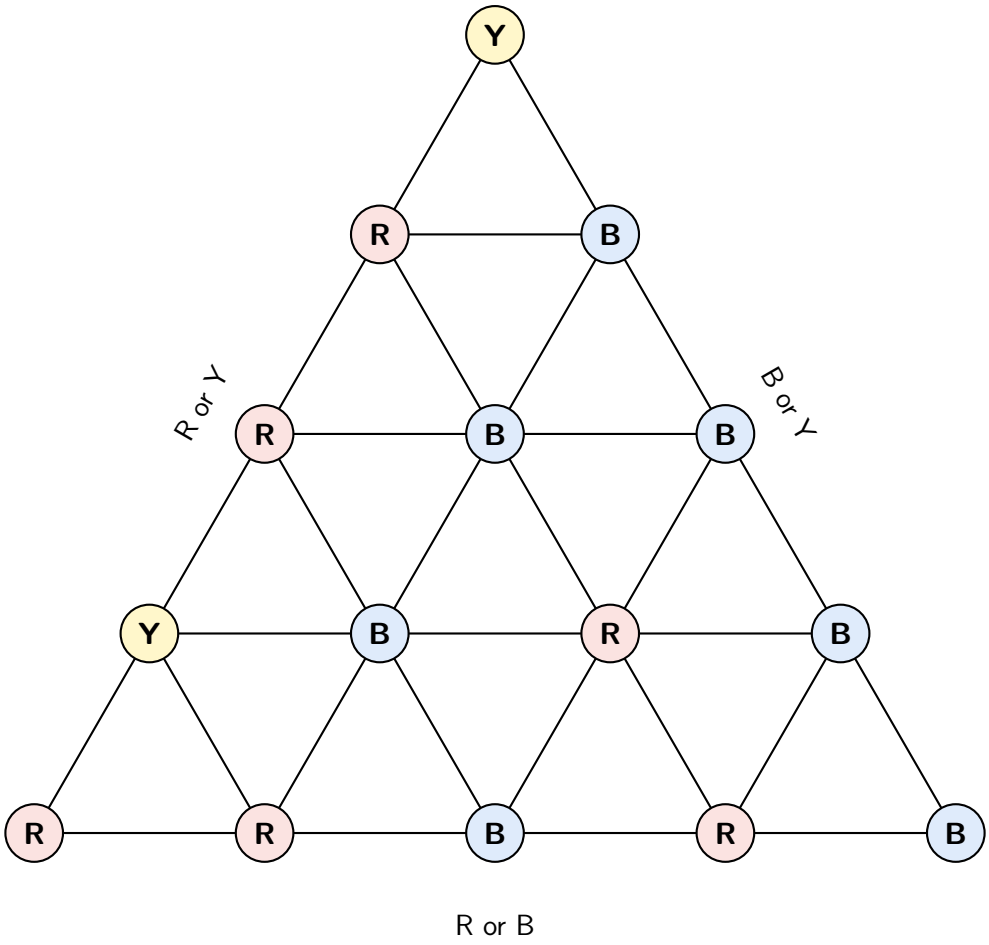
Problem 3: Fill these triangles with R, B, and Y in different ways, and mark the doors in each. Which fillings have exactly one door? Could a triangle have three doors?



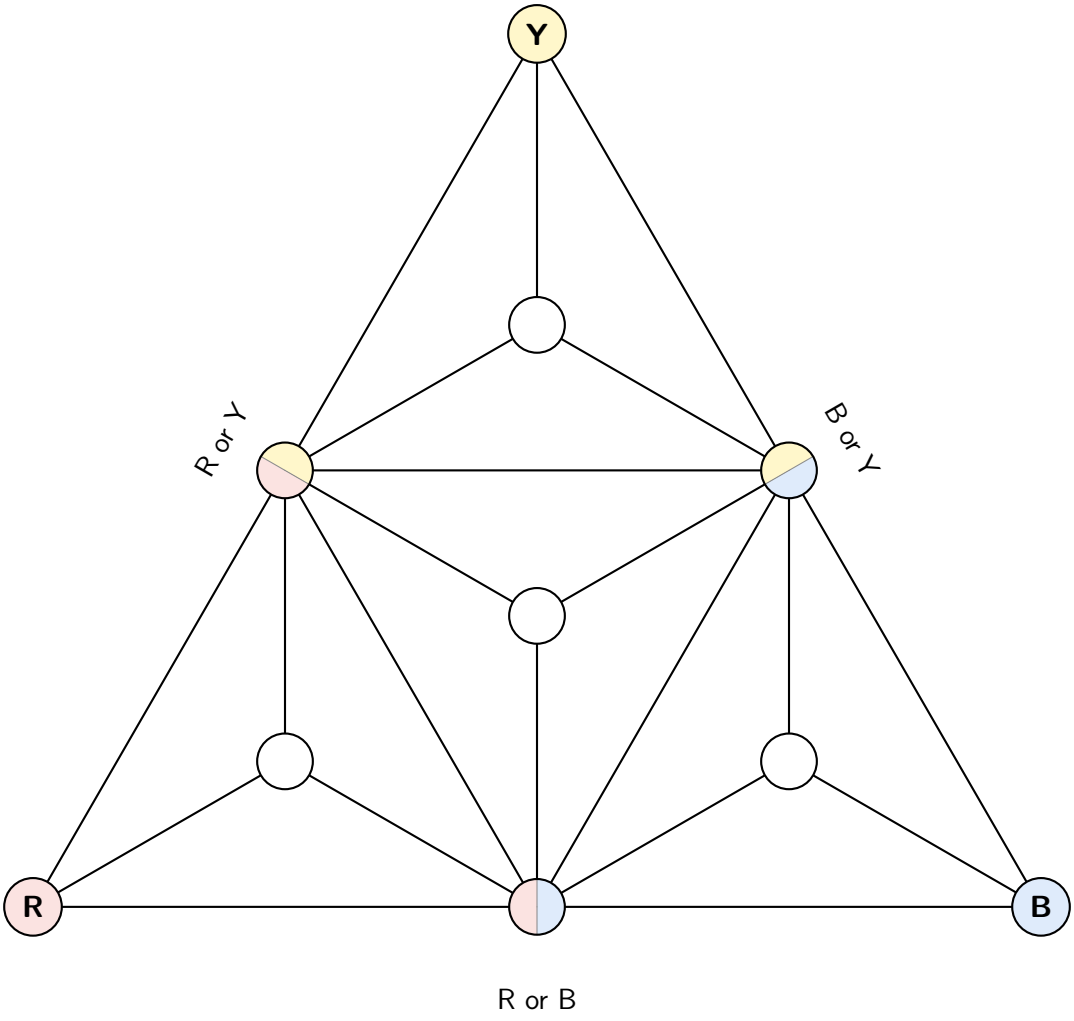
Problem 4: Mark every door. A route crosses through doors and uses no door twice. In a two-door triangle it must use both doors. Draw all the routes. Can every route from outside return outside?



Example: enter one thick door and leave through the other.
The dashed route crosses edges; it does not follow them.



Problem 5: Must the number of little triangles with R, B, and Y be odd? Explain your answer for every filling allowed by the side rules, on this board and on any division into little triangles that meet only at corners or along whole edges.



Problem 6: The starred dot may now use R, B, or Y. All other side rules still hold. Find a filling with no little triangle that has R, B, and Y. Explain why this does not contradict your answer to Problem 5.

